

Predictive Model Plan – Credit Delinquency

1. Model Logic (Generated with GenAI)

The objective of this model is to predict the probability of a customer becoming delinquent based on their financial and behavioral attributes. A logistic regression model is selected as the primary approach due to its simplicity and interpretability, with an optional advanced model (Gradient Boosted Trees) for improved accuracy.

Model Type:

- Primary: Logistic Regression
- Secondary: Gradient Boosted Trees

Key Input Features:

1. **Credit Utilization Ratio** – Indicates financial stress (higher = higher risk)
2. **Missed Payments (last 6 months)** – Strong predictor of delinquency
3. **Days Past Due** – Measures current repayment delay
4. **Income Level** – Reflects repayment capacity
5. **Number of Credit Inquiries** – Indicates credit demand and risk behavior

Model Workflow:

1. Data is collected and validated from customer records
2. Feature engineering is performed (e.g., utilization ratio, payment history trends)
3. Missing values are handled using median or rule-based imputation
4. Data is split into training and testing sets
5. Logistic regression model is trained to estimate delinquency probability
6. Predictions are generated as risk scores (0 to 1 probability)
7. Customers are classified into risk categories (Low, Medium, High)

Pseudocode:

Load dataset

Clean and preprocess data

Create features (utilization, missed payments, income band)

Split into train and test sets

Train logistic regression model

Predict probabilities on test data

Classify customers based on threshold

Evaluate performance using metrics

Model Summary

The model uses logistic regression to estimate the probability of customer delinquency based on financial behavior and credit usage patterns. It transforms input features into a risk score, enabling classification into risk categories. The approach ensures both interpretability and reliable prediction performance.

2. Justification for Model Choice

Logistic regression is well-suited for predicting credit delinquency because it provides high interpretability and transparency, which are essential in financial services. It allows Geldium to clearly understand how each feature impacts risk and ensures compliance with regulatory requirements. While more complex models like gradient boosting can improve accuracy, they often lack explainability, which is critical in lending decisions.

Additionally, logistic regression is easy to implement, fast to train, and scalable for large datasets. It balances performance and simplicity, making it ideal for real-world deployment where decisions must be explainable and auditable. This aligns with Geldium's need for a reliable, transparent, and efficient credit risk assessment system.

3. Evaluation Strategy

To ensure the model performs accurately and fairly, multiple evaluation metrics and checks will be used.

Performance Metrics:

- **Accuracy** – Measures overall correctness
- **Precision** – Ensures fewer false positives (avoiding mislabeling good customers)
- **Recall** – Captures most delinquent customers (reducing missed risks)
- **F1 Score** – Balances precision and recall
- **AUC-ROC** – Evaluates model's ability to distinguish between classes

Interpretation:

- High precision → reliable risk predictions
- High recall → better detection of risky customers
- High AUC → strong discrimination capability

Bias & Fairness Checks:

- Compare predictions across groups (e.g., income levels, regions)
- Detect proxy bias (e.g., location acting as sensitive attribute)
- Analyze false positive/negative rates across groups

- Apply fairness techniques like reweighting or equalized odds if needed

Ethical Considerations:

- Ensure model does not unfairly disadvantage specific groups
- Maintain transparency in decision-making
- Use explainability tools (e.g., SHAP)
- Include human oversight in final decisions

Final Note:

This model ensures a balance between accuracy, fairness, and interpretability, making it suitable for real-world financial risk prediction and aligned with Geldium's business goals.